

AI Video Analytics & ByteTrack Tracking Report

Video File: video_1785690472989_260990243.mp4 | Generated: 2026-08-02 22:38:46

1. Executive Summary

Automated AI vision tracking using YOLOv8 + ByteTrack was completed for **video_1785690472989_260990243.mp4**. The video was processed across **150 frames** (Duration: 6s, Resolution: 1920x1080). A total of **45 unique people** were assigned persistent tracking IDs. Maximum simultaneous occupancy reached **11** with an average occupancy of **7.6**. Crowd level was evaluated at **Low (22.0%)** based on a store capacity benchmark. Total footfall registered **19** (Entries: 13, Exits: 6) with an average detection confidence of **72.3%**.

2. Key Analytics Summary Table

Metric Parameter	Recorded Value	Benchmark / Context
Unique Tracked People	45	ByteTrack persistent unique tracking IDs
Current Occupancy	7	Visible tracked individuals in final frame
Maximum Occupancy (Peak)	11	Highest simultaneous tracked count
Minimum Occupancy	2	Lowest simultaneous tracked count
Average Occupancy	7.6	Mean tracked individuals per frame
Crowd Occupancy Rate	22.0%	Density Level: Low
Total Footfall Count	19	Entries: 13 Exits: 6
Average AI Confidence	72.3%	Range: 50.2% - 92.0%
Total Frames Processed	150	FPS: 25.0 Resolution: 1920x1080
Processing Execution Time	51.09 seconds	Real YOLOv8 + ByteTrack runtime

3. Object Class Statistics

Category	Count	Percentage
Person (Tracked)	45	100.0%

4. AI Insights & Recommendations

"Automated YOLOv8 + ByteTrack video analysis completed for video_1785690472989_260990243.mp4 (6s). Tracked 45 unique individuals across 150 frames with ByteTrack persistent ID mapping. Peak simultaneous occupancy reached 11 (Avg occupancy: 7.6). Crowd density evaluated at 22.0% (Low Status) relative to store capacity (50). Registered 13 entries and 6 exits with an average detection confidence of 72.3%."

- Unique shoppers tracked: 45. Ensure shelf layout maximizes engagement for peak traffic zones.
- Current crowd density level is Low (22.0%). Align staff allocation with peak occupancy windows.
- Recorded 13 entrance line crossings and 6 exit crossings.
- ByteTrack persistent tracking maintained across 150 frames with 72.3% average confidence.